Lab 9

Salary and Invoice System

https://github.com/mmujtaba25/cs-212

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Lab 9 - Part 1 - Employee

```java
package Lab9.Part1;

public abstract class Employee
{
    private static long lastID = 0;

    private final long employeeID;
    private String name;

    protected Employee(String name)
    {
        this.employeeID = lastID++;
        this.name = name;
    }

    public abstract double monthlyEarnings();

    /* GETTERS AND SETTERS */

    public long getEmployeeID() { return employeeID; }

    public String getName() { return name; }

    public Employee setName(String name)
    {
        this.name = name;
        return this;
    }
}
```

Lab 9 - Part 1 - FixedSalaryEmployee

```java
package Lab9.Part1;

public class FixedSalaryEmployee extends Employee
{
    private double baseSalary;
    private double allowance;

    protected FixedSalaryEmployee(String name) { super(name); }

    @Override
    public double monthlyEarnings() { return baseSalary + allowance; }

    /* GETTERS & SETTERS */

    public double getBaseSalary() { return baseSalary; }

    public FixedSalaryEmployee setBaseSalary(double baseSalary)
    {
        this.baseSalary = baseSalary;
        return this;
    }

    public double getAllowance() { return allowance; }

    public FixedSalaryEmployee setAllowance(double allowance)
    {
        this.allowance = allowance;
        return this;
    }
}
```

Lab 9 - Part 1 - HourlyEmployee

```java
package Lab9.Part1;

public class HourlyEmployee extends Employee
{
    private double hourlyRate = 15.0;
    private double hoursWorked;

    protected HourlyEmployee(String name) { super(name); }

    @Override
    public double monthlyEarnings()
    {
        return (hourlyRate * hoursWorked) * (hoursWorked > 40 ? 1.5 : 1);
    }

    /* GETTERS & SETTERS */

    public double getHourlyRate() { return hourlyRate; }

    public HourlyEmployee setHourlyRate(double hourlyRate)
    {
        this.hourlyRate = hourlyRate;
        return this;
    }

    public double getHoursWorked() { return hoursWorked; }

    public HourlyEmployee setHoursWorked(double hoursWorked)
    {
        this.hoursWorked = hoursWorked;
        return this;
    }
}
```

Lab 9 - Part 1 - InvoiceItem

```java
package Lab9.Part1;

public abstract class InvoiceItem
{
    private static long lastInvoiceNumber = 0;

    private final long invoiceNumber;
    private String description;

    public InvoiceItem(String description)
    {
        this.invoiceNumber = lastInvoiceNumber++;
        this.description = description;
    }

    public abstract double invoiceAmount();

    /* GETTERS AND SETTERS */

    public long getInvoiceNumber() { return invoiceNumber; }

    public String getDescription() { return description; }

    public InvoiceItem setDescription(String description)
    {
        this.description = description;
        return this;
    }
}
```

Lab 9 - Part 1 - Main

```java
package Lab9.Part1;

import java.util.ArrayList;

public class Main
{
    public static void main(String[] args)
    {
        final ArrayList<Employee> employees = new ArrayList<>();
        final ArrayList<InvoiceItem> invoices = new ArrayList<>();

        employees.add(new
FixedSalaryEmployee("Ali").setBaseSalary(3000).setAllowance(500));
        employees.add(new
HourlyEmployee("Talal").setHourlyRate(20).setHoursWorked(45));

        invoices.add(new
ProductInvoice("Laptop").setQuantity(2).setUnitPrice(1200));
        invoices.add(new
ServiceInvoice("Consulting").setHours(10).setRatePerHour(150));


        printTitle("Employees");

        for (Employee employee : employees)
            System.out.printf(
                    "%-16s %-24s Rs. %.2f%n",
                    employee.getName(),
                    employee.getClass().getSimpleName(),
                    employee.monthlyEarnings()
            );

        printTitle("Invoices");
        for (InvoiceItem invoice : invoices)
            System.out.printf(
                    "%-16s %-24s Rs. %.2f%n",
                    invoice.getDescription(),
                    invoice.getClass().getSimpleName(),
                    invoice.invoiceAmount()
            );
    }

    private static void printTitle(String Employees)
    {
        System.out.println();
        System.out.println(getCentered(Employees));
        System.out.println();
    }

    private static String getCentered(String text)
    {
        final int width = 54;
```

```
        int padding = (width - text.length()) / 2;
        return " ".repeat(Math.max(0, padding)) + text + " ".repeat(Math.max(0,
padding));
    }
}
```

Program Output

```

              Employees

Ali      FixedSalaryEmployee      Rs. 3500.00
Talal        HourlyEmployee       Rs. 1350.00

               Invoices

Laptop       ProductInvoice       Rs. 2400.00
Consulting   ServiceInvoice       Rs. 1500.00
```

Lab 9 - Part 1 - ProductInvoice

```java
package Lab9.Part1;

public class ProductInvoice extends InvoiceItem
{
    private int quantity;
    private double unitPrice;

    public ProductInvoice(String description) { super(description); }

    @Override
    public double invoiceAmount() { return quantity * unitPrice; }

    /* GETTERS & SETTERS */

    public int getQuantity() { return quantity; }

    public ProductInvoice setQuantity(int quantity)
    {
        this.quantity = quantity;
        return this;
    }

    public double getUnitPrice() { return unitPrice; }

    public ProductInvoice setUnitPrice(double unitPrice)
    {
        this.unitPrice = unitPrice;
        return this;
    }
}
```

Lab 9 - Part 1 - ServiceInvoice

```java
package Lab9.Part1;

public class ServiceInvoice extends InvoiceItem
{
    private double hours;
    private double ratePerHour;
    private double serviceFee;

    public ServiceInvoice(String description) { super(description); }

    @Override
    public double invoiceAmount() { return (hours * ratePerHour) + serviceFee; }

    /* GETTERS & SETTERS */

    public double getHours() { return hours; }

    public ServiceInvoice setHours(double hours)
    {
        this.hours = hours;
        return this;
    }

    public double getRatePerHour() { return ratePerHour; }

    public ServiceInvoice setRatePerHour(double ratePerHour)
    {
        this.ratePerHour = ratePerHour;
        return this;
    }

    public double getServiceFee() { return serviceFee; }

    public ServiceInvoice setServiceFee(double serviceFee)
    {
        this.serviceFee = serviceFee;
        return this;
    }
}
```

Lab 9 - Part 2 - Employee

```java
package Lab9.Part2;

public abstract class Employee implements Taxable
{
    private static long lastID = 0;

    private final long employeeID;
    private String name;

    protected Employee(String name)
    {
        this.employeeID = lastID++;
        this.name = name;
    }

    public abstract double monthlyEarnings();

    /* GETTERS AND SETTERS */

    public long getEmployeeID() { return employeeID; }

    public String getName() { return name; }

    public Employee setName(String name)
    {
        this.name = name;
        return this;
    }
}
```

Lab 9 - Part 2 - FixedSalaryEmployee

```
package Lab9.Part2;

public class FixedSalaryEmployee extends Employee
{
    private double baseSalary;
    private double allowance;

    protected FixedSalaryEmployee(String name) { super(name); }

    @Override
    public double monthlyEarnings() { return baseSalary + allowance; }

    @Override
    public double calculateTax() { return 0.12 * monthlyEarnings(); }

    /* GETTERS & SETTERS */

    public double getBaseSalary() { return baseSalary; }

    public FixedSalaryEmployee setBaseSalary(double baseSalary)
    {
        this.baseSalary = baseSalary;
        return this;
    }

    public double getAllowance() { return allowance; }

    public FixedSalaryEmployee setAllowance(double allowance)
    {
        this.allowance = allowance;
        return this;
    }
}
```

Lab 9 - Part 2 - HourlyEmployee

```java
package Lab9.Part2;

public class HourlyEmployee extends Employee
{
    private double hourlyRate = 15.0;
    private double hoursWorked;

    protected HourlyEmployee(String name) { super(name); }

    @Override
    public double monthlyEarnings() { return (hourlyRate * hoursWorked) *
(hoursWorked > 40 ? 1.5 : 1); }

    @Override
    public double calculateTax() { return 0.08 * monthlyEarnings(); }

    /* GETTERS & SETTERS */

    public double getHourlyRate() { return hourlyRate; }

    public HourlyEmployee setHourlyRate(double hourlyRate)
    {
        this.hourlyRate = hourlyRate;
        return this;
    }

    public double getHoursWorked() { return hoursWorked; }

    public HourlyEmployee setHoursWorked(double hoursWorked)
    {
        this.hoursWorked = hoursWorked;
        return this;
    }
}
```

Lab 9 - Part 2 - InvoiceItem

```java
package Lab9.Part2;

public abstract class InvoiceItem implements Taxable
{
    private static long lastInvoiceNumber = 0;

    private final long invoiceNumber;
    private String description;

    public InvoiceItem(String description)
    {
        this.invoiceNumber = lastInvoiceNumber++;
        this.description = description;
    }

    public abstract double invoiceAmount();

    /* GETTERS AND SETTERS */

    public long getInvoiceNumber() { return invoiceNumber; }

    public String getDescription() { return description; }

    public InvoiceItem setDescription(String description)
    {
        this.description = description;
        return this;
    }
}
```

Lab 9 - Part 2 - Main

```java
package Lab9.Part2;


import java.util.ArrayList;

public class Main
{
    public static void main(String[] args)
    {
        final ArrayList<Employee> employees = new ArrayList<>();
        final ArrayList<InvoiceItem> invoices = new ArrayList<>();

        employees.add(new
FixedSalaryEmployee("Ali").setBaseSalary(3000).setAllowance(500));
        employees.add(new
HourlyEmployee("Talal").setHourlyRate(20).setHoursWorked(45));

        invoices.add(new
ProductInvoice("Laptop").setQuantity(2).setUnitPrice(1200));
        invoices.add(new
ServiceInvoice("Consulting").setHours(10).setRatePerHour(150));

        printTitle("Employees");

        for (Employee employee : employees)
            System.out.printf(
                    "%-16s %-24s Rs. %.2f + Rs. %.2f tax%n",
                    employee.getName(),
                    employee.getClass().getSimpleName(),
                    employee.monthlyEarnings(),
                    employee.calculateTax()
            );

        printTitle("Invoices");
        for (InvoiceItem invoice : invoices)
            System.out.printf(
                    "%-16s %-24s Rs. %.2f + Rs. %.2f tax%n",
                    invoice.getDescription(),
                    invoice.getClass().getSimpleName(),
                    invoice.invoiceAmount(),
                    invoice.calculateTax()
            );
    }

    private static void printTitle(String Employees)
    {
        System.out.println();
        System.out.println(getCentered(Employees));
        System.out.println();
    }

    private static String getCentered(String text)
```

```
    {
        final int width = 69;
        int padding = (width - text.length()) / 2;
        return " ".repeat(Math.max(0, padding)) + text + " ".repeat(Math.max(0,
padding));
    }
}
```

Program Output

```

                  Employees

Ali      FixedSalaryEmployee      Rs. 3500.00 + Rs. 420.00 tax
Talal        HourlyEmployee       Rs. 1350.00 + Rs. 108.00 tax

                  Invoices

Laptop       ProductInvoice       Rs. 2400.00 + Rs. 144.00 tax
Consulting   ServiceInvoice       Rs. 1500.00 + Rs. 135.00 tax
```

Lab 9 - Part 2 - ProductInvoice

```java
package Lab9.Part2;

public class ProductInvoice extends InvoiceItem
{
    private int quantity;
    private double unitPrice;

    public ProductInvoice(String description) { super(description); }

    @Override
    public double invoiceAmount() { return quantity * unitPrice; }

    @Override
    public double calculateTax() { return 0.06 * invoiceAmount(); }

    /* GETTERS & SETTERS */

    public int getQuantity() { return quantity; }

    public ProductInvoice setQuantity(int quantity)
    {
        this.quantity = quantity;
        return this;
    }

    public double getUnitPrice() { return unitPrice; }

    public ProductInvoice setUnitPrice(double unitPrice)
    {
        this.unitPrice = unitPrice;
        return this;
    }
}
```

Lab 9 - Part 2 - ServiceInvoice

```java
package Lab9.Part2;

public class ServiceInvoice extends InvoiceItem
{
    private double hours;
    private double ratePerHour;
    private double serviceFee;

    public ServiceInvoice(String description) { super(description); }

    @Override
    public double invoiceAmount() { return (hours * ratePerHour) + serviceFee; }

    @Override
    public double calculateTax() { return 0.09 * invoiceAmount(); }

    /* GETTERS & SETTERS */

    public double getHours() { return hours; }

    public ServiceInvoice setHours(double hours)
    {
        this.hours = hours;
        return this;
    }

    public double getRatePerHour() { return ratePerHour; }

    public ServiceInvoice setRatePerHour(double ratePerHour)
    {
        this.ratePerHour = ratePerHour;
        return this;
    }

    public double getServiceFee() { return serviceFee; }

    public ServiceInvoice setServiceFee(double serviceFee)
    {
        this.serviceFee = serviceFee;
        return this;
    }
}
```

Lab 9 - Part 2 - Taxable

```java
package Lab9.Part2;

public interface Taxable
{
    double calculateTax();
}
```